PROJECT PLANNING PHASE

Script-Controlled ACL – Restrict Record Access Based on Field Value

1. Introduction

The project aims to develop a Script-Controlled Access Control List (ACL) mechanism that restricts access to records based on specific field values and user conditions. A custom script evaluates the requested record and the current user's information.

Access is granted only when the defined conditions are satisfied and the ACL script returns true. This approach provides dynamic and flexible record-level security compared with simple role-based access restrictions.

2. Problem Statement

Traditional access controls may provide the same level of access to all users having a particular role. However, organizations often need to restrict individual records based on their field values, ownership, department, location, or other conditions.

Therefore, a system is required to dynamically determine whether a user can access a particular record by evaluating predefined field values and user conditions.

3. Objectives

- Implement a script-controlled ACL mechanism.
- Restrict record access based on specific field values.
- Validate the current user's conditions before granting access.
- Provide record-level security.

- Prevent unauthorized users from viewing or modifying restricted records.
- Reduce the need for manually creating separate access rules for every condition.
- Maintain a secure and flexible access-control system.

4. Proposed System

The proposed system uses a custom ACL script to evaluate access requests.

When a user attempts to access a record:

1. The system identifies the requested record.
2. The ACL retrieves the required field values.
3. The system identifies the current user and relevant user attributes.
4. The custom script compares the record values with the user's conditions.
5. If the conditions are satisfied, the script returns true.
6. The user is granted access.
7. If the conditions are not satisfied, the script returns false and access is denied.

Basic Flow:

User → Record Request → ACL → Custom Script → Check Field Value + User Condition → Access Granted / Access Denied

5. Scope of the Project

Included:

- User authentication.
- Record-level access control.
- Field-value validation.

- User-condition validation.
- Script-based access decisions.
- Access granted/denied response.
- Testing of different user and record conditions.
- Basic security logging.

Not Included:

- Biometric authentication.
- Network-level security.
- Physical access control.
- Complete enterprise identity-management implementation.

6. Functional Requirements

The system should:

- Identify the logged-in user.
- Retrieve the requested record.
- Read the required field values.
- Evaluate the configured ACL script.
- Compare record fields with user information.
- Grant access when the condition evaluates to true.
- Deny access when the condition evaluates to false.
- Handle invalid or missing field values safely.
- Record access attempts where required.

7. Non-Functional Requirements

Security: The system must prevent unauthorized users from accessing restricted records.

Performance: ACL evaluation should be completed quickly without significantly affecting record access.

Reliability: The access-control mechanism should consistently produce the correct access decision.

Maintainability: The ACL script should be easy to modify when business requirements change.

Scalability: The system should support increasing numbers of users and records.

8. Example Use Case

Consider an organization where employee records contain a Department field.

Example:

Record 1 – Department: HR – User Department: HR – Access: Granted

Record 2 – Department: Finance – User Department: HR – Access: Denied

Record 3 – Department: IT – User Department: IT – Access: Granted

Condition:

Record Department == User Department

If the condition is true, the user can access the record. Otherwise, access is denied.

9. Project Modules

Module 1 – User Authentication

Identifies the currently logged-in user and retrieves required user information.

Module 2 – Record Management

Stores and retrieves records containing the fields that need to be protected.

Module 3 – ACL Configuration

Defines which records, operations, and fields require access restrictions.

Module 4 – Script Evaluation

Executes the custom ACL script and evaluates the specified conditions.

Module 5 – Access Decision

Grants or denies access according to the result of the ACL script.

Module 6 – Logging and Monitoring

Records important access attempts and ACL decisions for monitoring and auditing.

10. Technology Requirements

Software:

- Operating System: Windows/Linux
- Programming/Scripting Language: JavaScript or platform-supported scripting language
- Database: MySQL / PostgreSQL / platform database
- Web/Application Server: As required
- Development Environment: VS Code or platform-specific IDE
- Browser: Chrome/Edge/Firefox

Hardware:

- Computer/Laptop
- Minimum 4 GB RAM
- Minimum 20 GB available storage

- Internet/network connection where required

11. Project Development Plan

Phase 1 – Requirement Analysis: Identify project requirements.

Phase 2 – System Design: Prepare architecture and workflow.

Phase 3 – Database/Record Design: Design required tables and fields.

Phase 4 – ACL Design: Define access-control rules.

Phase 5 – Script Development: Develop the custom ACL script.

Phase 6 – Integration: Connect ACL with records and users.

Phase 7 – Testing: Perform functional and security testing.

Phase 8 – Security Validation: Verify access restrictions.

Phase 9 – Documentation: Prepare final project documentation.

Phase 10 – Deployment: Deploy the working system.

12. Testing Plan

Test Case 1: Valid user and matching field → Expected: Access Granted.

Test Case 2: Valid user but non-matching field → Expected: Access Denied.

Test Case 3: Unauthorized user → Expected: Access Denied.

Test Case 4: Missing field value → Expected: Access Denied.

Test Case 5: Multiple conditions satisfied → Expected: Access Granted.

Test Case 6: One required condition fails → Expected: Access Denied.

13. Security Considerations

- Access should be denied by default when required information is missing.
- Users should only receive the access necessary for their role and conditions.
- ACL scripts should be protected from unauthorized modification.
- Sensitive information should not be exposed through error messages.
- Access attempts should be logged where appropriate.
- ACL rules should be tested whenever they are modified.

14. Expected Outcome

At the end of the project, a working Script-Controlled ACL system will be developed. The system will dynamically evaluate record field values and user conditions before providing access.

Expected result:

Matching condition → Access Granted

Non-matching condition → Access Denied

This provides a flexible method for implementing record-level security.

15. Future Enhancements

- Multiple field-based access conditions.
- Department-based access control.
- Location-based access control.

- Record ownership rules.
- Time-based access restrictions.
- Approval-based access.
- Detailed audit reports.
- Administrative dashboards.
- Notifications for suspicious access attempts.
- Integration with enterprise identity-management systems.

16. Conclusion

The Script-Controlled ACL – Restrict Record Access Based on Field Value project provides a dynamic approach to record-level security. Instead of relying only on fixed roles, the system evaluates record fields and user conditions through a custom script.

By implementing this approach, organizations can control access more precisely and ensure that users can access records only when the defined security conditions are satisfied.